

Proposal Prep — LLM + Formal Methods for Trustworthy Agents

Reading notes on your supervisor's own work, plus the surrounding literature in code / law / medicine, oriented toward picking a proposal topic that **extends their program** rather than sitting beside it.

Compiled 28 Aug 2026. Every claim carries an arXiv ID or venue.

Benchmark figures are as reported by their authors and are not independently reproduced — re-check before putting any number in a written proposal.

Table of contents

1. [First: the files in this folder](#)
 2. [What the paper actually says](#)
 3. [The gaps *they* declare open](#)
 4. [What's already been done on those gaps](#)
 5. [Eight topics, ranked for this lab](#)
 6. [Background: the three domains](#)
 7. [How to run the meeting](#)
 8. [Bibliography](#)
-

1. First: the files in this folder

All three PDFs are **byte-identical** (197,121 bytes each) — one paper, saved three times:

CACM_LLM_FM.pdf

Position-Trustworthy-AI-Agents-Require-...-Formal-Methods

Position-Trustworthy-AI-Agents-Require-...-Formal-Methods

The paper: *Synergizing LLMs and Formal Methods for Neural-Symbolic Reasoning* Jin Song Dong, Yufan Cai, Zhe Hou, Jing Sun, Hongshu Wang, Yedi Zhang, Xinyue Zuo *Journal of the ACM*, 2025. (Also circulating as a Position paper — [OpenReview](#).)

The group: NUS (Dong, Cai, Wang, Zhang, Zuo) + Griffith (Zhe Hou) + Auckland (Jing Sun). This is the PAT / model-checking lineage — Dong is a co-author of the original PAT toolkit (CAV 2009).

△ Note the file naming: you may be missing the actual *Position* paper, if it's a separate document from the JACM article. Worth checking with your prof that you have both.

2. What the paper actually says

Read this section before the meeting. Being able to restate their paradigm in your own words is worth more than any topic idea.

2.1 The core argument

- LLM hallucination is **not a bug to be trained away** — it is mathematically inherent to the probabilistic objective (they cite Xu et al., *Hallucination is Inevitable*, [arXiv:2401.11817](#)).
- Formal methods give rigorous guarantees but have a steep learning curve and poor accessibility.
- Therefore: **LLMs propose, formal methods dispose.**

Their motivating example is deliberately humble: a 3-digit combination puzzle that older LLMs get wrong with fluent-but-invalid reasoning. Claude 4 translated it into SMT-LIB and **Z3 solved it in 0.01 seconds on a laptop**. The point is not that the puzzle is hard; it's that the *right division of labour* makes it trivial.

2.2 The paradigm

LLMs explore the design space by proposing candidate artifacts; FM backends enforce a certificate-based acceptance boundary grounded in explicit assurance targets.

The key concept is the **certificate** — machine-checkable evidence produced by the FM backend: a localized conflict explanation (unsat core), a violating execution trace (counterexample), or an undischarged proof obligation.

They deliberately choose **external** FM (constrain / check / guide LLM outputs without touching model internals) over tightly-integrated approaches like Logical Neural Networks or Scallop — arguing internal integration is "possible but prohibitively resource-intensive."

Architecture components: a Task Automation LLM (orchestrator), an **Auto-formalization LLM**, a Designing LLM, a Coding LLM, RAG for domain knowledge, and FM engines (SMT solvers, theorem provers, model checkers) whose outputs feed a repair loop.

2.3 The five design rules — memorise these

Rule	Content
1. Assurance target	State the artifact that must be correct and the properties defining correctness. Write the acceptance claim as <i>"this output is acceptable because <property> holds for <artifact>."</i> If you can't state it precisely, design drifts toward optimising fluency instead of correctness

2. Backend FM choice	Match backend to property class and domain of quantification. SMT → constraint satisfiability/consistency, returns unsat cores . Model checking → behavioral/temporal properties, returns counterexample traces , first-class safety/liveness/fairness. Theorem proving / refinement → semantic, unbounded correctness via proof obligations . Lighter-weight fallbacks: type systems, abstract interpretation, contract checking
3. LLM–FM interface	Define the contract: what the LLM may propose (candidates, decompositions, repairs) and what FM must return on failure. Treat formal failures as typed signals, not generic "try again" — organize repair around channels: inconsistency / counterexample / unsatisfied obligation
4. Integration pattern	Match coupling to dominant failure mode. Decision/compliance → consistency-guarded deliberation . System behaviors → counterexample-guided refinement . Component synthesis → proof-obligation-guided synthesis . Combine across layers: certify high-level model first, then constrain lower-level synthesis
5. Acceptance guard	State explicitly what certificate gates acceptance and what it certifies. Stronger certificates → higher iteration cost, narrower applicability. FM is the correctness filter, not an optional add-on ; keep clean separation between candidate generation and certificate-based validation

2.4 The three case studies

#	Name	Direction	FM backend	Certificate	Referen
1	Legal AI (L4L)	FM4LLM	SMT (Z3)	Unsat core	Chen, Cai, Ho Dong — arXiv:2511

2	PAT-Agent	LLM4FM	Model checking (PAT/CSP#)	Counterexample trace	Zuo et al., AS
3	Trustworthy AI Coder	"They are one"	Refinement calculus + ATP	Discharged proof obligations	Cai et al., PO 2025

Case study 1 — Legal (FM4LLM: formal methods improving the LLM). Each statutory clause is parsed into a typed rule `Rule(c) := <Actor, Action, Cond, Penalty>` and compiled into a first-order implication. A **Prosecutor LLM** (accusatory stance) and an **Attorney LLM** (exculpatory stance) independently extract fact predicates and cited clauses. A neutral **Auto-formalizer LLM** turns their evidence into SMT-LIB; Z3 checks it. On UNSAT, the **minimal unsatisfiable core** is extracted, mapped back to the original text, and used to instruct the offending agent to drop contradictory clauses. A Judge LLM renders the verdict. Their worked example (a Chinese narcotics case) yields 2 years + ¥10,000 against a ground truth of 1 year 2 months + ¥11,000. Reported as highest accuracy among SOTA methods.

Case study 2 — PAT-Agent (LLM4FM: LLMs lowering the FM entry barrier). Four transformations: $T_{plan}: L_NL \rightarrow \Pi \rightarrow P, T_{gen}: P \rightarrow M, V: M \times Q \rightarrow \{MATCH, MISMATCH\} \times C, T_{repair}: M \times C \times R \rightarrow M$. A Planning LLM extracts constants/variables/actions/guards and produces a generation plan; a Coding LLM emits CSP#; PAT verifies; counterexamples drive a Repair Agent that fixes **only the faulty region** rather than regenerating. Reported: **99.6% requirement-translation accuracy**; Claude 3.7 Sonnet produces near-perfect CSP# syntax first try; **3–4 repair iterations typically suffice**. Their closing line is the pitch: *"automakers can now verify their designs without the excuse of 'But we do not have formal methods engineers.'"*

Case study 3 — Trustworthy AI Coder. GPT-5 writes a plausible `sqrt` function that is **wrong** — `abs(x_next - x) <= e` does not imply `abs(x_next2 - N) <= e`. Their fix uses Morgan's **refinement calculus**: start from a first-order logic spec, apply correctness-preserving transformations (assignment introduction, guard strengthening, loop introduction), with the LLM predicting which

refinement law to apply, instantiating it, and emitting the induced proof obligation; an ATP discharges the POs. Failures return the failing proof state, and the LLM revises, switches rule, or backtracks.

The killer result to quote: LLM pass rates *decrease* as you add test cases (bugs remain). Their method's pass rate is **constant regardless of test count, because the programs are proven correct**. Remaining failures are mostly *theorem-prover timeouts*, not incorrect code.

3. The gaps *they* declare open

This is where your proposal should live. Section 6 of the paper — "Mitigating Auto-formalization Risk" — is effectively a research agenda handed to you.

3.1 The stated central problem

Semantic under-specification. "Natural language omits assumptions, scopes, and constraints, so an auto-formalizer can produce a logically consistent but **intent-misaligned** specification that still passes solver checks and yields **certificates for the wrong target**."

And, crucially:

"Formal methods act as **verifiers of consistency, not arbiters of semantic truth**. Bridging the informal–formal gap therefore remains an open research challenge."

Restated as the hole in their own architecture: Design Rule 5 puts an acceptance guard on the *output artifact*. **There is no acceptance guard on the specification itself**. The autoformalizer is the one unverified component in a system whose entire selling point is verification.

3.2 Their explicit future-work list

From §6 and §7, they name these as the way to close the gap:

1. **N-version auto-formalization** — generate multiple independent formalizations (different LLMs or prompt/decoding variants), filter with lightweight formal checks (well-formedness/type constraints, satisfiability, non-triviality, consistency against fixed background constraints). **Material disagreement is treated as a high-risk signal and surfaced explicitly rather than silently resolved.**
2. **Router + specialists** — a system-level mixture-of-experts dispatching to domain- and logic-specific formalizers, keeping the same certificate-producing backend as the gate.
3. **Typed intermediate representations + constrained decoding** — LLM emits a type-checked schema compiled *deterministically* into the target formalism.
4. **Scenario-based semantic validation** — round-trip paraphrases and auto-generated "**should / should-not**" scenarios checked against both original intent and the formal solver.

3.3 Their named application frontiers

"compliance analysis, protocol verification, cyber-physical safety, and synthesis of critical components"

Plus the Lean ecosystem: Mathlib, and **CSLib — The Lean Computer Science Library** (Barrett et al., [arXiv:2602.04846](#)), which they flag as enabling "results beyond what either could readily do alone."

3.4 The gap they *don't* name — and you should

Medicine is missing. Their case studies cover law (SMT), systems (model checking), and code (theorem proving). Healthcare appears once, in the introduction's list of LLM application areas, and never again. Given that clinical guidelines are *already* a classical formal-methods target (see §4.3), this is the most natural fourth case study in the paper — and it is unclaimed.

4. What's already been done on those gaps

I checked the literature against each of their future-work items so you don't propose something already published. **This section is your novelty defence.**

4.1 N-version autoformalization — ⚠ partially scooped

"Neurosymbolic Auditing of Natural-Language Software Requirements" ([arXiv:2605.13817](#)) does almost exactly the N-version idea: it samples multiple independent SMT formalizations of the same requirement and compares them with **bidirectional SMT equivalence checking**, treating persistent disagreement as evidence that the requirement admits more than one plausible reading.

What this means for you: the mechanism is published, in the *software-requirements* domain. Still open: (a) doing it where formalization targets are **statutes or clinical guidelines**, not requirements; (b) turning disagreement into a **quantitative risk certificate** rather than a binary flag; (c) the router+specialists half, which nobody has built. **Do not propose plain N-version-for-requirements.** Cite this paper and differentiate explicitly.

4.2 Autoformalization fidelity — an active, adjacent field

Work	ID	Relevance
------	----	-----------

ConsistencyCheck	—	859 expert-annotated items; binary "does this formal statement preserve the semantics?" Even human experts make semantic errors in up to 38.5% of cases. Devastating number for any human-confirmation-based mitigation — and their §6 assumes exactly that
Verus-SpecGym	2605.26457	Agentic environment for spec autoformalization. Finding: specification autoformalization is a distinct bottleneck — models that write correct code for a problem often fail to write a faithful spec for the same problem
Faithful Autoformalization via Roundtrip Verification and Repair	2604.25031	Their round-trip-paraphrase idea, already implemented
Evaluating Autoformalization Robustness via Semantically Similar Paraphrasing	2511.12784	Paraphrase-invariance as a fidelity test
ReForm: Reflective Autoformalization	2510.24592	Prospective bounded sequence optimization
Faults in Our Formal Benchmarking	2606.29493	Dataset defects in Lean theorem-proving benchmarks

Towards
Autoformalization of
LLM-generated
Outputs for
Requirement
Verification

2511.11829

Closest to the SE-
requirements framing

Takeaway: round-trip verification and paraphrase robustness are *taken*. The **certificate** framing — producing a machine-checkable artifact that bounds formalization risk, and gating acceptance on it — is **not**, and it is the natural extension of their own Design Rule 5.

4.3 Clinical guidelines + model checking — a 15-year-old literature with a stalled bottleneck

This is the important discovery. There is a substantial pre-LLM body of work:

- *Adopting model checking techniques for clinical guidelines verification* — **Artificial Intelligence in Medicine** (Elsevier)
- *Using model checking for critiquing based on clinical guidelines* — **Artificial Intelligence in Medicine**, 2008 — casts guideline *critiquing* in temporal logic; compares actual treatment derived from real patient data against the guideline as "ideal actions"
- *Analyzing Clinical Practice Guidelines Using a Decidable Metric Interval-Based Temporal Logic* (Springer)
- Reviews of care-pathway computerization document that **manual formalization is the adoption blocker** for process-oriented health information systems

Read that against PAT-Agent. The classical literature says: *model checking clinical guidelines works, but nobody can afford to write the models*. PAT-Agent says: *we automated writing the models, at 99.6% requirement-translation accuracy*. **These two literatures have not been joined.** That is Topic 01.

4.4 Their own follow-on work — already in flight

- **Event-B Agent: Towards LLM Agent for Formal Model Synthesis and Repair** — Wang, Zuo, Sun, Li, Ait Ameur, Dong. **FSE 2026**, [arXiv:2605.17475](#). Same group, extending the LLM4FM line to Event-B.
- **On Verifiable Legal Reasoning: A Multi-Agent Framework with Formalized Knowledge Representations** ([arXiv:2509.00710](#)) — an independent group, same idea as L4L. Check overlap.
- **ATA: A Neuro-Symbolic Approach to Autonomous and Trustworthy Agents** ([arXiv:2510.16381](#)).

△ Ask whether an Event-B / Lean / CSLib direction is already assigned to another student before you commit to anything in that lane.

5. Eight topics, ranked for this lab

Ranked by **fit to their program × novelty × feasibility**. Every one is framed as *LLM proposes* → *FM certifies*, because that is the lab's thesis.

Rank 01 — The missing fourth case study: FM4Medicine

Auto-formalized clinical guidelines with counterexample-guided care-plan checking

Their paper has case studies in law, systems, and code. **Medicine is absent**. Meanwhile a 15-year AI-in-Medicine literature established that clinical practice guidelines can be model-checked in temporal logic — and stalled because **manual formalization was unaffordable**. PAT-Agent solved exactly that bottleneck for CSP#. Join them.

Assurance target (DR1)	<i>"This care plan is acceptable because it satisfies the temporal properties derived from guideline G."</i>
-------------------------------	--

Backend (DR2)	Model checking — the properties are inherently temporal/behavioral (sequencing, contraindication windows, escalation ordering, monitoring intervals). Their own Rule 2 selects it
Certificate (DR5)	A counterexample trace showing the concrete sequence of clinical actions that violates the guideline
Pattern (DR4)	Counterexample-guided refinement — same as PAT-Agent
Build	Planning LLM + Coding LLM pipeline from guideline prose → CSP# (or Event-B, reusing the group's newest work). Check retrospective care plans, or plans produced by a medical agent on MedAgentBench / AgentClinic, against the formalized guideline
Measure	Formalization accuracy vs. clinician annotation; violation detection rate; how many guideline violations by an LLM medical agent are caught that no benchmark metric catches
Why it lands	Reuses their tool, their paradigm, and their design rules; opens a new domain; the "excuse of no formal methods engineers" line applies <i>verbatim</i> to hospitals. And it converts medical agent evaluation from accuracy-scoring to compliance-certification , which is what regulators will actually demand
Risk	Guideline prose is more ambiguous than statutes — which is itself the §6 auto-formalization problem, so it feeds Topic 02. Mitigate by starting with a narrow, well-structured guideline (sepsis bundles, anticoagulation dosing, antibiotic stewardship)

Prior art to cite and differentiate from: the AI-in-Medicine model-checking papers (pre-LLM, manual formalization) and PAT-Agent (LLM formalization, non-medical domain). Nobody has bridged them.

Rank 02 — An acceptance guard for the auto-formalizer

Turning formalization risk into a machine-checkable certificate

Their Design Rule 5 gates the *output artifact* with a certificate. **The specification itself is ungated** — the one unverified component in a verification-first architecture. §6 names this precisely: a formalizer can yield "certificates for the wrong target."

Build	A layered fidelity guard: (i) N-version formalization with bidirectional equivalence checking; (ii) auto-generated should / should-not scenarios discriminating candidate formalizations; (iii) a quantitative risk score compiled into a certificate that gates acceptance alongside the solver's
Measure	Detection of intent-misaligned-but-satisfiable specs. Build the eval set by deliberately injecting semantic under-specification (dropped scope, missing precondition, wrong quantifier) and measuring catch rate vs. round-trip and paraphrase baselines
Why it lands	It is the group's own §7 future work, and it makes their whole paradigm <i>sound</i> rather than merely useful. Publishable as infrastructure regardless of outcome
Risk	⚠ Novelty pressure. N-version SMT-equivalence is done for SE requirements (2605.13817); round-trip is done (2604.25031); paraphrase robustness is done (2511.12784). Differentiate on three axes: the certificate/risk-quantification framing, the domain (statutes and guidelines, not code requirements), and integration into an end-to-end acceptance guard rather than a standalone metric. Say this explicitly in the meeting — showing you know what's taken is worth more than the idea itself

Sharpest supporting fact: ConsistencyCheck reports **human experts miss semantic errors in up to 38.5% of cases.** Their §6 mitigation is "the user manually confirms the formal specification." That mitigation is empirically unreliable. Lead with this.

Rank 03 — Do certificates actually help the human?

An empirical study of unsat cores and counterexample traces as expert-facing explanations

The entire paradigm assumes a human validates the spec and consumes the certificate. **Nobody has tested whether certificates improve expert decisions.** The reliance-calibration literature is actively discouraging: standard interventions (numeric confidence, feature-based explanations) **do not robustly improve calibration and can worsen over-reliance** ([arXiv:2502.13321](#)), and in one clinical study physicians overestimated an AI's accuracy by ~30 percentage points.

Build	A controlled study. Same underlying task (legal consistency check or care-plan check), three conditions: no explanation / natural-language LLM explanation / formal certificate (unsat core, counterexample trace). Measure whether experts correctly accept correct outputs and reject incorrect ones
Measure	Appropriate reliance — sensitivity <i>and</i> specificity, not raw accuracy. Whether certificates cause over-trust ("it's formally verified, so it's right") even when the spec is wrong
Why it lands	Cheap, fast, no GPUs. It de-risks the group's entire program by testing its human-facing assumption, and it's the kind of result that gets cited by everyone in LLM+FM. A negative result is <i>more</i> interesting than a positive one
Risk	Needs human subjects (ethics approval, participant recruitment). Start small — even 15–20 domain-trained participants gives a publishable pilot. Law students are more accessible than clinicians

Rank 04 — Certifying the *process*, not just the artifact

Temporal-logic acceptance guards over agent tool-call traces

Their certificates gate an output: a verdict, a model, a program. But an agent is a *trace of actions*, and in law and medicine the **procedure is itself legally mandated** — conflict checks before advice, contraindication checks before prescribing, consent before disclosure. Lift the acceptance guard from the artifact to the run.

Build	A specification language for professional-procedure constraints + a runtime monitor over the agent's tool-call trace that can block or roll back a violating action. Extends AgentLTL (2607.02599); the specification/verification/enforcement survey (2608.14590, 38 studies 2022–2026) is your related-work map
Measure	Violation rate with/without enforcement; the utility cost of enforcement (over-blocking) — this trade-off is exactly their DR5 "stronger certificates raise iteration cost"
Why it lands	A clean, principled extension of their Design Rule 4/5 into agentic settings, in the domains where "usually correct" is not an acceptable standard
Risk	Formalising real professional obligations needs domain experts. Overlaps Rank 01 — could be the second half of the same thesis

Rank 05 — Proof obligations as free process-reward signal

Bridging their POPL work to the agentic-RL literature

The SWE-agent field is spending enormous effort *learning* step-level reward models (SWE-RM, 2512.21919, 30B MoE; SWE-TRACE, 2604.14820) because outcome-only reward is too sparse for long trajectories. **The refinement calculus already emits a dense, objective, per-step correctness signal for free: the discharged proof obligation.**

Build	Use PO discharge/failure as ground-truth process supervision to train or evaluate a process reward model. Then test the interesting direction: does a PRM trained on formally-verified traces transfer to domains with <i>no</i> verifier?
Measure	PRM accuracy against PO ground truth vs. LLM-as-judge baselines; downstream test-time-scaling gains
Why it lands	Connects the lab's formal work to the hottest thread in agent training, and gives the SWE-RL community something it can't generate itself — certified process labels. Good joint-venue potential (PLDI/POPL ↔ ICLR/NeurIPS)
Risk	Refinement-calculus traces are a narrow distribution; transfer may simply fail. Frame transfer as the research question so a negative result still publishes

Rank 06 — Is Design Rule 2 actually right?

Empirically validating property-class → FM-backend selection

Design Rule 2 asserts that SMT suits satisfiability/consistency, model checking suits behavioral/temporal, theorem proving suits unbounded semantic correctness. It is stated as expert guidance and **never measured**. Test it.

Build	A benchmark of tasks spanning property classes, run through all three backends, holding the LLM and prompt fixed. Report where the rule holds, where it breaks, and what the crossover costs are
Measure	Success rate, iteration count, wall-clock, and failure mode by backend (timeout vs. wrong answer vs. spec mismatch)

Why it lands	Converts your supervisor's design heuristic into a measured result — flattering, useful, and squarely within their frame. It is also the LLM+FM analogue of the coding-benchmark critique (2606.17799) about confounded components
Risk	Engineering-heavy (three toolchains). Low intellectual risk, moderate labour. A good <i>first</i> paper before a bigger thesis

Rank 07 — L4L beyond statute: what happens when law isn't a rule?

Testing the limits of the compile-a-clause-to-a-typed-rule assumption

L4L works because Chinese statutory law compiles cleanly into `<Actor, Action, Cond, Penalty>`. Common-law reasoning — precedent, analogy, balancing tests, open-textured standards like "reasonable" — does not.

Build	Apply the L4L architecture to a common-law or Indian-law corpus. Characterise where the typed-rule abstraction holds, degrades, and fails. Where it fails, what is the right weaker certificate — a <i>partial</i> consistency check over the formalisable fragment?
Measure	Coverage (fraction of provisions formalisable), accuracy on the formalisable fragment, and honest failure analysis of the rest
Why it lands	Directly extends the group's own case study, and the legal-agent surveys explicitly flag jurisdictional skew — datasets are heavily Chinese, common-law coverage is thin . If you have any access to Indian case law, this is your unfair advantage

Risk	Needs legal annotation. The honest outcome may be "much of common law doesn't formalise" — which is a real finding but a harder sell. Pre-agree with your supervisor that a negative characterisation counts
-------------	--

Rank 08 — Unsat core as a principled abstention trigger

Replacing confidence scores with certificates in high-stakes agents

Medical-agent safety work (MedAbstain, 2601.12471, EACL 2026) finds models often fail to abstain when uncertain — and worse, that measured abstention is partly a **prompt artifact rather than genuine uncertainty** (2507.16199). Confidence scores are the wrong instrument. A solver saying *UNSAT*, and here is the minimal conflicting set is a **principled, inspectable** abstention trigger.

Build	An agent whose action space includes escalate-to-human, triggered by certificate failure rather than by token probability. Evaluate on MedAgentBench write-actions (the risky ones) or a legal drafting task
Measure	Risk–coverage curves vs. confidence-based abstention. Control for the prompt-artifact confound
Why it lands	Unifies the lab's certificate framing with the agent-safety literature; gives abstention a semantics instead of a threshold
Risk	Only fires where the task is formalisable at all — so it's really an extension of Rank 01 or 04 rather than a standalone thesis

Quick selection guide

If...	Pick
-------	------

You want the clearest "new contribution to my supervisor's program" story	01
You want to work on the problem they explicitly called open	02
You have little compute and want a fast first publication	03 or 06
You want to connect the lab to mainstream ML/agent research	05
You have access to Indian legal data	07
You want the biggest thesis (multi-paper)	01 → 04 as one arc, or 02 → 01

6. Background: the three domains

Condensed. Full detail and all sources are in the [visual briefing](#). Useful for related-work sections and for showing breadth in the meeting.

6.1 The frame that connects them

The shared bottleneck is verification, not capability. 2023–2025 was the capability phase — can the model write the patch, find the case, name the diagnosis. Frontier models now clear the static exams. 2026 is **verification under autonomy**. Each domain has a different oracle, and its strength predicts the domain's speed:

Domain	Oracle	Strength	Consequence
Code	Tests, compilers, runtime	Cheap, automatic	Fastest progress; large-scale RL possible. But incomplete — passing tests ≠ correct

Law	Citation to authority; adversarial process	Checkable in form, not substance	Citation <i>existence</i> nearly solved; doctrinal correctness has no automatic checker
Medicine	Outcome (eventually), guidelines (now)	Delayed months; ethically un-explorable	No RL on real patients → simulation → fidelity gap

This is exactly the argument for the lab's program. A formal certificate is a *constructed* oracle for domains that lack a natural one. Say it that way in the meeting.

6.2 Code — what to know

- **Standard pipeline:** localisation → repair → validation → selection. Survey: 2512.22256 (242 studies).
- **RL is the centre of gravity.** SWE-RM (2512.21919) trains an execution-free reward model: Qwen3-Coder-Flash 51.6% → 62.0%, Max 67.0% → 74.6% on SWE-bench Verified.
- **The benchmark revolt.** 2606.17799 argues benchmarks *collapse model, harness, and environment into one score*, where any component can move results by a full model generation. SWE-bench Pro (2509.16941): 1,865 tasks, **best pass@1 still 23.3%**. Contamination is real — models reproduce gold patches from the task ID alone.
- **Security:** indirect prompt injection succeeds **4.7% @1 attempt, 33.6% @10, 63.0% @100** (2605.17634).
- **The uncomfortable fact:** the METR RCT found experienced OSS developers were **19% slower** with early-2025 AI tools while estimating they were 20% faster.
- ⚠ Extremely crowded. Do not compete on SWE-bench score from a university lab.

6.3 Law — what to know

- **Taxonomy survey:** 2601.06216. Five practice areas; three pillars (external grounding via RAG and tool use; procedural orchestration; multi-layer verification modelled on firm peer review).
- **The landmark negative result:** Magesh et al., *Journal of Empirical Legal Studies* 2025 (Stanford RegLab/HAI) — first pre-registered evaluation of commercial AI legal research tools. Vendors marketed RAG as "hallucination-free"; measured misleading output **~1 in 6**. The residual failure isn't inventing cases — it's **citing real authority that doesn't support the claim**, or that's been overruled. → *Verifying a citation exists is a lookup. Verifying it supports the claim is an open problem. This is the strongest possible motivation for the SMT approach in L4L.*
- **Benchmarks:** LegalAgentBench (2412.17259, 300 tasks / 17 corpora / **37 tools**, reports intermediate progress rate), LegalBench, LEXam, PLawBench (2601.16669), DLawBench (2606.13931), MASLegalBench.
- **Known weakness:** agents are **brittle under perturbation** — alter a material fact and reasoning often fails to adjust, implying pattern-matching over doctrine.
- **Stated gap:** jurisdictional skew, heavily Chinese. Common-law coverage thin. → Rank 07.

6.4 Medicine — what to know

- **AgentClinic** (2405.07960): reformulating static MedQA as interactive sequential encounters drops diagnostic accuracy to **below one-tenth** in some settings. The "LLM passes USMLE" headlines measured recall of a fully-specified vignette; real medicine is **information acquisition under uncertainty and cost**.
- **MAI-DxO / SDBench** (2506.22405): 304 NEJM CPC cases with **per-test dollar cost**. 80% accuracy vs 20% generalist-physician baseline (85.5% accuracy-maximised); **–20% cost vs physicians, –70% vs bare o3**. *Caveats to raise yourself:* physicians had no colleagues, textbooks, or search; CPC cases are selected for rarity.
- **MedAgentBench** (2501.14654, now *NEJM AI*): FHIR-compliant

virtual EHR, 300 physician-authored tasks, 100 patients, 700k+ data elements. Frontier models **56–72%**. **Key asymmetry: retrieval queries are easy; write/modify actions are hard** — which is precisely where a certificate-gated acceptance guard belongs.

- **Abstention:** MedAbstain (2601.12471) — models often fail to abstain when uncertain. Confound: abstention is partly a prompt artifact (2507.16199).
 - **Automation bias:** physicians overestimated an AI's accuracy by **~30 percentage points**.
 - **The unaudited assumption:** every interactive result rests on an LLM playing a cooperative, articulate, internally consistent patient. Real patients are none of those.
-

7. How to run the meeting

Open by proving you read the paper

Restate the paradigm in one sentence — *"LLMs propose candidate artifacts, formal methods gate acceptance with machine-checkable certificates, and formal failures become typed feedback that drives repair"* — then name the three case studies and their three backends. This takes twenty seconds and completely changes the conversation.

Then make one observation

Pick the sharpest:

- *"Medicine is the missing case study, and clinical guidelines were already a model-checking target before LLMs — the blocker was manual formalization, which PAT-Agent solves."*
- *"Design Rule 5 gates the artifact, but nothing gates the specification. The autoformalizer is the only unverified component in a verification-first architecture."*
- *"Section 6's mitigation is human confirmation of the spec, but*

ConsistencyCheck reports experts miss semantic errors up to 38.5% of the time."

Bring three topics, not eight

- **Ambitious / new domain:** Rank 01
- **Their stated open problem:** Rank 02
- **Fast and cheap:** Rank 03 or 06

Say explicitly which is which. Supervisors respond well to a student who has already thought about scope risk.

Questions to ask them

1. **Is the Event-B / Lean / CSLib direction already assigned to another student?** (Wang et al. FSE 2026 shows the group is actively expanding there — you need to know what's taken.)
2. **Do we have a clinical or legal collaborator?** This decides between Ranks 01/07 and 02/05/06.
3. **Is there a domain the group wants to enter next?** Their §7 names compliance analysis, protocol verification, cyber-physical safety, synthesis of critical components — medicine is conspicuously absent, and it's worth asking whether that's deliberate.
4. **How much of the PAT / PAT-Agent toolchain can I build on directly?** Reusing it makes Rank 01 dramatically more feasible.
5. **Target venue and timeline?** ASE/FSE/POPL are their venues; that shapes whether you write a tool paper, a benchmark, or an empirical study.
6. **Do you want the auto-formalization gap attacked generally, or inside one domain?** Their §7 implies general; a thesis is usually better inside one.

Objections to have an answer ready for

Objection	Answer
-----------	--------

<i>"N-version formalization is already published."</i>	Yes — 2605.13817, for software requirements , as a binary ambiguity flag. Open: statutes/guidelines as targets, risk quantification , and integration as an acceptance guard. Name the paper before they do
<i>"Isn't guideline formalization already done?"</i>	The AI-in-Medicine literature did it manually , and manual formalization is documented as the adoption blocker. PAT-Agent removes exactly that blocker. The two literatures have never been joined
<i>"Will formal methods scale to real clinical text?"</i>	Probably not fully — which is why the honest scope is a narrow, well-structured guideline first, and why coverage is a reported metric rather than an assumption
<i>"Do benchmark numbers matter?"</i>	The METR RCT says not automatically. That's an argument for certificate-based and human-facing evaluation (Rank 03), not against the field

8. Bibliography

8.1 The group's own work — read all of these first

Ref	Work	Venue
—	Synergizing LLMs and Formal Methods for Neural-Symbolic Reasoning — Dong, Cai, Hou, Sun, Wang, Zhang, Zuo	J. ACM 2025 · OpenReview · <i>the PDF in this folder</i>

2511.21033	Towards Trustworthy Legal AI through LLM Agents and Formal Reasoning (L4L) — Chen, Cai, Hou, Dong	Case study 1. Four stages: statute knowledge building → dual fact/statute extraction → solver-centric adjudication → judicial rendering
—	PAT-Agent: Autoformalization for Model Checking — Zuo, Zhang, Wang, Cai, Hou, Sun, Dong	ASE 2025 , pp. 2122–2133. Case study 2
—	Automated Program Refinement: Guide and Verify Code LLM with Refinement Calculus — Cai, Hou, Sanan, Luan, Lin, Sun, Dong	POPL 2025 . Case study 3
2605.17475	Event-B Agent: LLM Agent for Formal Model Synthesis and Repair — Wang, Zuo, Sun, Li, Ait Ameer, Dong	FSE 2026 . The group's newest direction
—	PAT: Towards Flexible Verification under Fairness — Sun, Liu, Dong, Pang	CAV 2009. The underlying toolkit

8.2 Foundations they build on

Ref	Work
2401.11817	Xu, Jain, Kankanhalli — Hallucination is Inevitable: An Innate Limitation of LLMs . <i>The paper's philosophical premise</i>
2405.06624	Dalrymple et al. — Towards Guaranteed Safe AI

—	Wing — Trustworthy AI , CACM 64(10), 2021
—	de Moura & Bjørner — Z3: An Efficient SMT Solver , TACAS 2008
—	Morgan — Programming from Specifications , 1990. <i>The refinement calculus behind case study 3</i>
—	Hoare — Communicating Sequential Processes , CACM 1978. <i>CSP# ancestor</i>
2602.04846	Barrett et al. — CSLib: The Lean Computer Science Library . Flagged in their future work
—	Yang et al. — LeanDojo , NeurIPS 2023. Their example of <i>tight</i> integration
—	Li, Huang, Naik — Scallop , PLDI 2023
—	Riegel et al. — Logical Neural Networks , arXiv:2006.13155
—	Pan et al. — Logic-LM , EMNLP Findings 2023

8.3 Autoformalization fidelity — the Rank 02 related-work core

ID	Work	Why it matters
2605.13817	Neurosymbolic Auditing of Natural-Language Software Requirements	⚠ N-version SMT-equivalence, already published. Cite and differentiate
2605.26457	Verus-SpecGym: Agentic Environment for Evaluating Specification Autoformalization	Spec autoformalization is a distinct bottleneck from code generation
2604.25031	Faithful Autoformalization via Roundtrip Verification and Repair	Their round-trip idea, implemented

2511.12784	Evaluating Autoformalization Robustness via Semantically Similar Paraphrasing	Paraphrase-invariance as fidelity test
2510.24592	ReForm: Reflective Autoformalization	Prospective bounded sequence optimization
2511.11829	Towards Autoformalization of LLM-generated Outputs for Requirement Verification	Closest SE-requirements framing
2606.29493	Faults in Our Formal Benchmarking	Dataset defects in Lean benchmarks
2506.10903	Beyond Gold Standards: Epistemic Ensemble of LLM Judges for Formal Mathematical Reasoning	Ensemble judging
—	ConsistencyCheck	859 expert-annotated items; experts make semantic errors up to 38.5% of the time

8.4 Agent safety, specification, enforcement — Rank 04 core

ID	Work
2608.14590	Toward Safe LLM Agents: A Survey of Specification, Verification, and Enforcement (38 studies, 2022–2026)
2607.02599	AgentLTL: Trace Verification for Procedural Compliance in Tool-Using Agents
2508.03665	A Design-by-Contract-Inspired Neurosymbolic Layer for Trustworthy Agent Design

2511.09008	A Neurosymbolic Approach to Natural Language Formalization and Verification
2510.16381	ATA: Neuro-Symbolic Autonomous and Trustworthy Agents
2605.10325	Verifiable Process Rewards for Agentic Reasoning

8.5 Legal

ID	Work
JELS 2025	Magesh et al. — Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. <i>The anchor negative result</i>
2601.06216	LLM Agents in Law: Taxonomy, Applications, Challenges
2509.00710	On Verifiable Legal Reasoning: Multi-Agent Framework with Formalized Knowledge Representations. Check overlap with L4L
2412.17259	LegalAgentBench — 300 tasks, 17 corpora, 37 tools, reports intermediate progress rate
2605.21071	Fine-grained Claim-level RAG Benchmark for Law
2505.03970	A Reasoning-Focused Legal Retrieval Benchmark
2504.01840	LRAGE: Legal RAG Evaluation Tool
2606.30906	Investigating Multi-Agent Deliberation in Law
2602.12056	LawThinker: Deep Research Legal Agent

8.6 Medical — Rank 01 core

ID	Work
----	------

—	Adopting model checking techniques for clinical guidelines verification — <i>Artificial Intelligence in Medicine</i> (Elsevier)
—	Using model checking for critiquing based on clinical guidelines — <i>Artificial Intelligence in Medicine</i> , 2008
—	Analyzing Clinical Practice Guidelines Using a Decidable Metric Interval-Based Temporal Logic — Springer
PMC3197986	Computerization of workflows, guidelines, and care pathways: implementation challenges. Documents manual formalization as the blocker
2405.07960	AgentClinic — accuracy collapse from static to sequential
2506.22405	Sequential Diagnosis with Language Models (SDBench / MAI-DxO)
2501.14654	MedAgentBench — FHIR virtual EHR, now <i>NEJM AI</i>
2601.12471	Knowing When to Abstain: Medical LLMs Under Clinical Uncertainty (EACL 2026)
2507.16199	LLM Abstention Can Be a Prompt Artifact. The confound to control for
2607.07761	Aligning Clinical Needs and AI Capabilities: Survey on LLMs for Medical Reasoning

8.7 Code agents — Rank 05/06 context

ID	Work
2512.22256	Agentic Software Issue Resolution with LLMs: A Survey (242 studies)
2512.21919	SWE-RM: Execution-free Feedback for SWE Agents
2604.14820	SWE-TRACE: Rubric Process Reward Models + Test-Time Scaling

2606.17799	Position: Coding Benchmarks Are Misaligned with Agentic SE
2509.16941	SWE-Bench Pro — best pass@1 23.3%
2605.17634	AI Agents May Always Fall for Prompt Injections
metr.org	METR RCT — 19% slower, perceived 20% faster

8.8 Human factors — Rank 03 core

ID	Work
2502.13321	Adjust for Trust: Mitigating Trust-Induced Inappropriate Reliance on AI Assistance
2510.26518	Human-AI Complementarity: A Goal for Amplified Oversight
PMC11612524	Facilitating Trust Calibration in AI-Driven Diagnostic Decision Support — physicians overestimated accuracy by ~30 points

Sources are the PDF in this folder, public arXiv preprints, and peer-reviewed articles located by literature search. Reported figures are as stated by their authors and have not been independently reproduced.